

- *1. The reduced row echelon form of a system of linear equations is given. Write the system of equations corresponding to the given matrix. Use x , y , and z as variables. Determine whether the system is consistent or inconsistent. If it is consistent, give the solution.

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -4 \\ 0 & 1 & 9 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

What equation does the first row represent?

(Type an equation.)

What equation does the second row represent?

(Type an equation.)

What equation does the third row represent?

(Type an equation.)

Determine whether the system is consistent or inconsistent. If it is consistent, give the solution. Select the correct choice below and, if necessary, fill in the answer box(es) within your choice.

- ☐ A. The system is consistent. The solution is $(\text{}, \text{}, \text{})$.
(Simplify your answers.)
- ☐ B. There are infinitely many solutions. The solution can be written as $\{(x,y,z) | x = \text{}, y = \text{}, z \text{ is any real number}\}$.
(Simplify your answers. Type expressions using z as the variable.)
- ☐ C. There are infinitely many solutions. The solution can be written as $\{(x,y,z) | x = \text{}, y \text{ is any real number}, z \text{ is any real number}\}$.
(Simplify your answer. Type an expression using y and z as the variables.)
- ☐ D. The system is inconsistent.

2. Solve the following system.

$$\begin{cases} 3x - 2y + 5z = 22 \\ x - 2y - 2z = -3 \\ 2x - 2y = 0 \end{cases}$$

Select the correct choice below and, if necessary, fill-in answer boxes to complete the choice.

- ☐ A. There is exactly one solution. The solution is $x = \text{}$, $y = \text{}$, and $z = \text{}$.
(Simplify your answers. Type integers or simplified fractions.)
- ☐ B. There are infinitely many solutions of the form $x = \text{}$ and $y = \text{}$, where z is any real number.
(Type expressions using z as the variable.)
- ☐ C. There is no solution.

*3. Solve the following system.

$$\begin{cases} \frac{5}{4}x - \frac{4}{3}y + \frac{5}{4}z = -5 \\ \frac{5}{16}x + \frac{3}{4}y - \frac{3}{16}z = 6 \\ \frac{3}{4}x - y + \frac{3}{4}z = 6 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- ☐ A. The system has a single solution. The solution set is $\{(\text{ }, \text{ }, \text{ })\}$.
(Simplify your answer.)
- ☐ B. There are infinitely many solutions and the equations are dependent. The solution set is $\{(x, y, z) | \text{ }\}$.
- ☐ C. The system is inconsistent. The solution set is the empty set.

*4. Solve the system.

$$-9x - 3y + 4z - 4 = 0$$

$$6x + 6y + 2z - 3 = 0$$

$$-3x - 4y + 6z - 14 = 0$$

The solution set is $\{(\text{ }, \text{ }, \text{ })\}$.
(Type an integer or a simplified fraction.)

5. This system does not have a unique solution. Solve the system if possible.

$$\begin{cases} x - y + 2z = -2 \\ 4x + 3y + 7z = -9 \\ x + 6y + z = -3 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. There are infinitely many solutions. If z is allowed to be any real number, then $x = \text{ }$ and $y = \text{ }$.
(Type expressions using z as the variable. Use integers or fractions for any numbers in the expression.)
- ☐ B. There is no solution.

6. This system does not have a unique solution. Solve the system, if possible.

$$7x - 5y - 9z = 10$$

$$6x - 5y - 8z = 23$$

$$x - z = 0$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. There are infinitely many solutions. If z is allowed to be any real number, then $x = \text{ }$ and $y = \text{ }$.
(Type expressions using z as the variable.)
- ☐ B. There is no solution.

- *7. Solve the following system of equations using matrices (row operations). If the system has no solution, say that it is inconsistent.

$$\begin{cases} 3x + y - z = \frac{1}{4} \\ 2x - y + z = 1 \\ 6x + 2y = \frac{5}{2} \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box(es) in your choice.

- ☐ A. The solution is (, ,).
(Simplify your answers.)
- ☐ B. There are infinitely many solutions. The solution can be written as $\{(x,y,z) | x = \text{}, y = \text{}, z \text{ is any real number}\}$.
(Simplify your answers. Type expressions using z as the variable.)
- ☐ C. There are infinitely many solutions. The solution can be written as $\{(x,y,z) | x = \text{}, y \text{ is any real number}, z \text{ is any real number}\}$.
(Simplify your answer. Type an expression using y and z as the variables.)
- ☐ D. The system is inconsistent.

- *8. Solve the following system of equations using matrices (row operations). If the system has no solution, say that it is inconsistent.

$$\begin{cases} x - y + z = 1 \\ 4x - y + 4z = 4 \\ 3x + y + 3z = 3 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box(es) in your choice.

- ☐ A. The solution is (, ,).
(Simplify your answers.)
- ☐ B. There are infinitely many solutions. The solution can be written as $\{(x,y,z) | x = \text{}, y = \text{}, z \text{ is any real number}\}$.
(Simplify your answers. Type expressions using z as the variable.)
- ☐ C. There are infinitely many solutions. The solution can be written as $\{(x,y,z) | x = \text{}, y \text{ is any real number}, z \text{ is any real number}\}$.
(Simplify your answer. Type an expression using y and z as the variables.)
- ☐ D. The system is inconsistent.